

Systematic Review of AI-Based Personalized Diet and Activity Recommendation System for Diabetic Individuals

Abstract. Diabetes is a significant and growing public health problem not only globally but also in Sri Lanka. Sri Lanka has very traditional dietary practices and limited access to personalized management tools to understand this problem. This systematic review focuses on AI-based personalized diet and activity recommendation systems for diabetic patients in Sri Lanka, with a focus on technologies adaptable to environments like Sri Lanka. Using PRISMA guidelines, we analyzed 38 studies (2014–2025) across some domains. AI-driven food recognition, portion estimation, wearable sensor-based health monitoring, glucose prediction, digital twin modelling, and federated learning. Results indicate that wearable sensors and predictive analytics enable continuous glucose monitoring and forecasting, while deep learning models (CNNs, vision transformers) and augmented reality (AR) have a great capacity for precise food recognition and portion estimation. New ideas, such as digital twins and federated learning, make it possible to give nutrition advice that is very personal and confidential. There are, however, big gaps in the recognition of meals and patterns that are specific to certain cultures (such as Sri Lankan dishes). This review says that a full AI-powered system that uses these technologies could assist people with diabetes in managing their condition better by giving them personalized advice on diets and activities that fit their culture. This would help keep blood sugar levels stable and boost health in general.

Keywords: Calorie Intake Estimation, Digital Twin with federated learning, Food Recognition, Glucose Level Estimation, Portion Size Estimation, Wearable Sensor-Based Monitoring

1 Introduction

Diabetes is a growing and serious health problem around the world, not just in Sri Lanka. In Sri Lanka, between the ages of 18 and 29, 3.6% of people have diabetes [1], and 23% of adults are aged 18 and above [1]. In the Western Province, 27% of adults over the age of 20 have the disease [2]. The frequency rises to 55% for people between the ages of 60 and 69 [2]. Studies show that people in their 20s are more likely to get diabetes [3].

It's very important to keep track of your daily exercise and food intake if you have diabetes. But most of the systems we have currently simply keep track of food or activity, and they don't always give a whole answer. In Sri Lanka, there are many cultures that have many culturally appropriate ways of eating; this could lead to bad calorie intake to physical activity usages. Sometimes suggestions can be wrong, and it can be

hard to find those suggested meals, and due to limitations of cultural knowledge, that portion size estimation can go wrong.

This review focuses on an AI-driven personalized system capable of recognizing Sri Lankan cuisine, assessing its nutritional value, monitoring user activities through smartphone and smart band sensors, and recommending simple home workouts to mitigate these issues, showing the progress via the new technologies like digital twin and federated learning. This treatment is meant to help people with diabetes.

2 Review Methodology

The approach employed for this systematic review adheres to the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) standards.

2.1 Search Strategy

MDPI, ScienceDirect, PubMed, Springer, and IEEE Xplore were the five main data bases chosen to do a thorough search of the literature. Recent research in AI-driven healthcare and diabetes management employed academic papers published from 2014 to 2025. We used keywords to locate the best articles for this literature review study. I utilized Boolean operators to combine the terms "food recognition," "portion estimation," "calories estimation," "wearable sensors for diabetes," "glucose prediction," "digital healthcare," and "personalized nutrition" to find the publication that was most relevant to my investigation.

2.2 Inclusion and Exclusion Criteria

Inclusion criteria covered open-access, peer-reviewed studies involving ML or DL techniques applied to dietary analysis, activity monitoring, or glucose prediction. Exclusion criteria removed papers not in English, non-experimental studies, editorials, and research focused exclusively on hardware

After the filtering procedure, the number of articles selected from each source is shown in **Table 1**.

Table 1. Number of Articles found from each resources

Source	Number of articles extracted
MDPI	26
PubMed	2
Springer	4
ScienceDirect	2
IEEE Journals	4
Total	38

2.3 Study Selection Process

This systematic review found 247 papers by searching through resource databases. There were 179 different things to look at after getting rid of the duplicates. A careful look at the titles and abstracts got rid of 123 papers that didn't fit with the goals of the research. Next, we carefully examined the other 66 full-text papers to see if they were suitable for our review. We discovered that 28 studies were unsuitable due to insufficient data, methodological issues, or irrelevance to the study's objective. In the end, 38 publications were picked to be part of the qualitative synthesis. (See Fig. 1. PRISMA diagram of literature identification and selection process)

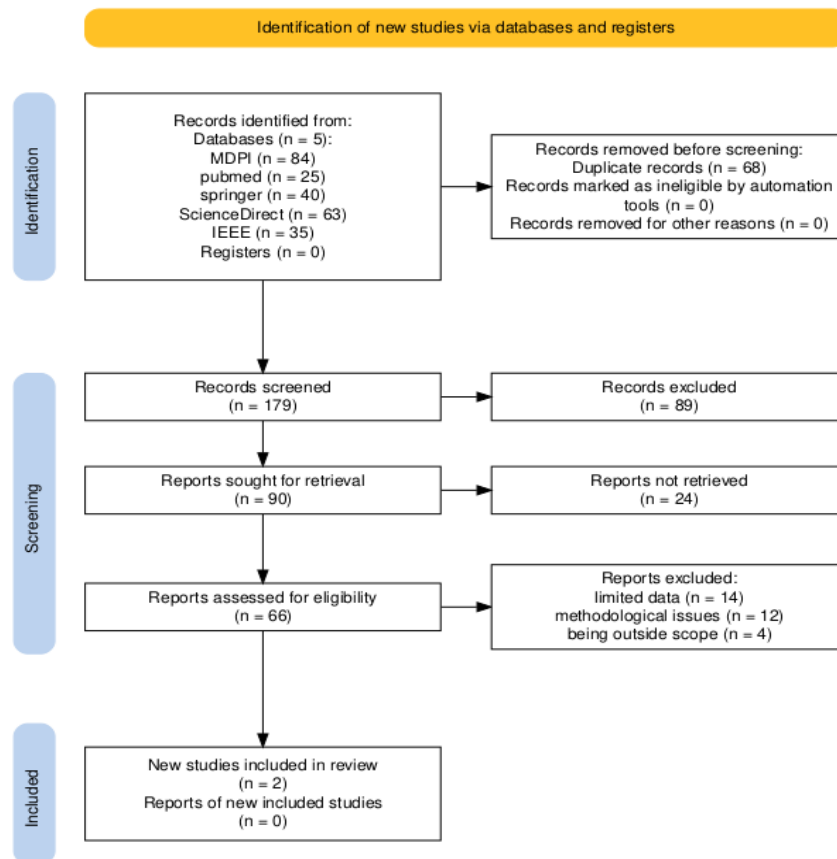


Fig. 1. PRISMA diagram of literature identification and selection process

3 Classification of Existing Researches

3.1 Food Recognition and Classification Systems

Recent research explains that artificial intelligence and computer vision are rapidly advancing automated dietary and nutrition assessment tools that give a personalized approach for health management, enabling objective estimation of food intake, nutrient composition, and health-related variables from images with reduced human bias. Deep learning models such as ResNet and vision transformers can directly predict dish mass and nutrient content from food images, improving real-time digital health tools for nutritional monitoring [4]. Image-based methods that combine color and depth (RGB-D) image data using object detection networks like Mask R-CNN excellently enhance food volume and intake estimation accuracy compared with manual measures and traditional processes [5].

AI-based dietary assessment systems can match or potentially exceed human estimation accuracy for calories and volume, although variability in databases and reporting limits broad meta-analysis, eliminating fixed diet patterns [6]. As modern research shows, the usage of LLM for food recognition provides good classification when compared to other traditional methods [7]. AI helps with personalized diet recommendation systems for people with diabetes, chronic disease management, and malnutrition monitoring by combining wearable sensors and mobile phones for remote monitoring, image recognition, and machine learning algorithms [8]. However, it stresses the importance of ethics and clinical validation before widespread use [9]. All of these study demonstrates that AI could help make better guesses about nutrition and make dietary changes that are right for each person.

3.2 Portion Estimation

For public health, nutritional evaluation, and nutrition monitoring, it's crucial to be able to accurately guess how much food you need. Self-reported methods that have been used for a long time typically include mistakes, which has led to the usage of image-based and augmented reality (AR) technology. Early studies used mobile phones to capture food images and estimate portion sizes, forming the basis for modern digital and AR-based dietary assessment systems [10]. Later research increased the accuracy by using 3D reconstruction, depth sensing, and object scaling methods, enabling more reliable estimation of food volume from single images or monocular cameras [11], [12], [13].

More recent studies show that AR is effective for both real-time portion estimation and portion size education. Mobile AR tools, such as virtual portions and AR overlays, help users better understand portion sizes and calorie usage based on them compared to

traditional time consuming guides [14], [15]. Research studies also show that simple visual-based guidance in AR significantly improves the accuracy of food portion estimation, especially for unfamiliar foods [16], [17]. In general, AR-based food portion estimation uses picture analysis, 3D modeling, and user interaction to make it less likely that individuals will guess the wrong portion size of food. These methods are superior for real-world use, when keeping track of nutrition accurately is important for dealing with problems like malnutrition and health disorders related to diet.

3.3 Wearable Sensors and Continuous Health Monitoring

Recent research indicates that wearable sensor technologies are increasingly vital for ongoing health monitoring, particularly for individuals with diabetes. These devices collect real-time data such as physical activity, heart rate, glucose levels, and energy expenditure, which helps doctors and patients better understand daily health patterns. An earlier research study shows that combining ambient and wearable sensors can accurately estimate daily energy usage in older adults [18], which is useful because physical activity strongly affects blood glucose control in a better manner. Zahedani et al. showed that digital health applications integrating wearable sensor data with behavior patterns can significantly improve metabolic health, highlighting the value of long-term monitoring rather than occasional clinic visits [19]. Some research focused specifically on diabetes and proposed predictive glucose monitoring systems using wearable sensors, allowing early detection of abnormal glucose trends before serious health issues occur to the diabetes patient [20]. Shi et al. did a research study focusing on wearable monitoring by integrating various physiological and behavioral information to improve nutritional monitoring for patients managing diabetes [21]. Prior studies validated that wearable technology may accurately assess nutritional intake, endorsing their therapeutic application [22]. The literature suggests that wearable sensors enable continuous, non-invasive, and individualized health monitoring for diabetes control. They are especially helpful for people with busy lives and in places where regular hospital-based monitoring is hard to do since there aren't enough resources.

3.4 Glucose Prediction and Predictive Analytics

Wearable sensors and artificial intelligence are commonly employed for continuous glucose monitoring and forecasting to enhance diabetes management. Initial studies indicated that glucose levels and hypoglycemic incidents can be accurately predicted using data from continuous glucose monitoring devices, with nonlinear and machine-learning models demonstrating superior performance compared to conventional linear approaches [23]. Recent study emphasizes deep learning methodologies, utilizing intricate datasets and realistic instructions to assist developers in creating precise glucose prediction systems suitable for real-world application [24]. Non-invasive glucose prediction with multi-modal wearable sensors has enhanced patient comfort and facilitates

monitoring in free-living conditions, thereby simplifying patients' lives and optimizing daily diabetes management [25]. Advanced models have also been developed for predicting blood glucose levels in children with Type 1 diabetes, demonstrating good results in managing complicated and changing glucose patterns, which helps doctors make quick treatment decisions. [26]. Smart home environments that combine IoT and wearable sensors are also a relatively new way to monitor glucose levels and anticipate health in older people, making it easier to care for them [27]. In general, these studies suggest that integrating wearable sensors to predict glucose levels with AI allows for early intervention, individualized care, and continuous monitoring. This is especially useful in healthcare settings with limited resources and for people with busy lives.

3.5 Digital Twin Concepts and AI-Driven Personalized Nutrition

Digital twins are exact virtual copies of a patient that combine the patient's real clinical data, exercise (from wearables) readings, and lifestyle information for the simulation process of the patient's real life. These virtual duplicates let clinicians safely guess and model how a person's health might change. These twins can help them make better choices without putting the patient's life at danger. Recent studies indicate that AI-generated digital twins can facilitate personalized healthcare by accurately modeling disease state, forecasting treatment responses, and evaluating the impacts of lifestyle modifications, thereby conserving time and resources for patients. [28], [29]. Digital twins in diabetes treatment can combine information on a patient's food, blood sugar, exercise, and medication use to create individualized future body forecasts. This will assist patients in avoiding unhealthy foods and better managing their illness [30]. The concept is effectively applied in various medical fields, where deep learning models analyze clinical testing data to predict a patient's response to specific treatments [28], [31]. All things considered, digital twins provide a better method of precision medicine that enables medical professionals to foresee health issues, maximize therapies, and develop really individualized treatment programs for every patient.

3.6 Federated and AI-Driven Personalized Nutrition

Federated learning is an AI method that lets models learn from data from many hospitals or other places without sharing raw patient data. From that way, it is protecting the privacy of that particular patient while improving model performance. This is particularly important in healthcare applications such as diabetes patient management and medical imaging environments. Where data privacy is important and regulatory constraints limit centralized data collection. Recent work shows that federated learning enables collaborative medical data mining while maintaining data security, making it suitable for large-scale, multi-institutional healthcare systems [32], [33]. When combined with continuous glucose monitoring and personalized glucose forecasting models, federated learning can improve hypoglycemia prediction and individualized diabetes care without exposing patient data in raw format to particular AI systems. This approach is also relevant for large screening systems, such as AI-based early warning tools for diabetic predictions, where data from multiple centers can be used safely to

improve detection accuracy of the particular AI system; this could lead to better prediction of the users [33].

4 Comparative Analyses of Existing Approaches

The table below shows the comparison of existing studies with the key differences and similarities across datasets, methodologies, tools, and evaluations. This comparison will help to identify research trends, commonly used technologies, and performance benchmarks that will help to find out the research gaps of current research. (See Table 2)

Table 2. Systematic Review chart for comparison approach

Author	year	Algorithm/ Method	Dataset	evaluation	Limitations	Ref.
R.Bianco <i>et al</i>	2025	ResNet-50, ResNet-101, InceptionV3, ViT-B-16	Nutri- tion5k	MAPE: 6.2% (mass), 6.4% (energy), 12.3% (fat), 33.3%(pro- tein), 40.2%(car- bohydrate)	2D imagery without depth limits volume accuracy	[4]
K. A. Nfor <i>et al.</i>	2025	CNN (ResNet- 50), ViT-B16, Grad-CAM, LIME	Food101, Vireo- Food-172, CNFOOD- 241	~93.7% on Vireo- Food172, ~91.3% on Food101	Limited regional food diver- sity	[34]
Y. Han <i>et al.</i>	2023	CNN + Depth Prediction + RGB-D Fusion	Nutri- tion5k	Nutrition estima- tion errors 10– 22%	Limited regional food diver- sity	[35]
S.-T. Cheng <i>et al.</i>	2025	LLaMA-3, Multi- view Stereo	CNFOOD- 241 Food101 dataset	Top-1 Classifica- tion Accuracy: 91% on CNFOOD-241 Top-1 Classifica- tion Accuracy: 80% on Food101	High vol- ume estima- tion error	[7]
L. Bossard <i>et al.</i>	2014	Random Forest, SVM, CNN baseline	Food-101 Dataset	Accuracy 50.76%	Inferior to modern DL models	[36]
D. Lee and S. Kwon,	2024	Mask R-CNN (RGB-D)	Custom RGB-D	Region detection accuracy (IoU) of Mask R-CNN: 94.1%, Volume	Sensitive to food stack- ing	[5]

				estimation error up to 2.2%		
Z. Chen et al.	2025	Convolutional Neural Network	Food-101, Vireo-Food-172	Top-1 acc. ~90% (CETCH Food 101), 89.86% (Vireo Food – 172), 80.61% (UEC Food-100)	High computational cost	[37]
L. Boyd et al.	2024	CNNs (VGG16, ResNet50)	Food-101, UEC Food-100	Val. acc. 0.79	No transformer exploration	[38]
A. Mariappan et al.	2009	Image segmentation (pre/post meal)	Manual images	Accuracy 70% (with good light)	No AR, lighting sensitivity	[10]
W. Jia et al.	2022	3D bowl reconstruction	Custom	~10% overestimation	Liquid bias	[11]
G. Vinod et al.	2024	3D object scaling from 2D	Simple-Food45	Avg. error 17.6%	Requires prior 3D models	[12]
X. Xu et al.	2021	Monocular + cross-domain adaptation	Custom	MAPE 11.47%	Limited free-living data	[13]
V. Pini et al.	2023	AR overlay guidance	Grocery images	Qualitative improvement	No direct calorie estimation	[15]
V. Maksimenko et al.	2024	Eye-tracking + AR	Custom	Avg. error ~7%	Environment-dependent	[16]

5 Discussion

This systematic review shows the current use of artificial intelligence in personalized diabetes care, highlighting both progress and significant gaps, particularly in countries like Sri Lanka, which has diverse cultural patterns, varying appropriate eating habits for individuals, and limited resources. A lot of AI systems can now accurately identify food and guess how much of it is in a picture. Most of these systems, on the other hand, are trained on datasets that don't have South Asian or Sri Lankan meals. Because of this, individuals often don't know how to appropriately identify diverse cultural meals or ways of eating in their area. Similarly, tools that use augmented reality to estimate portions are not designed for local tableware and traditional ways of eating.

The review also discusses the use of wearable devices to predict blood glucose levels and the patient's daily activity. These devices can collect activity data continuously and

help patients manage their daily calorie prediction and help to manage diabetes more easily. However, most studies test these diabetes management systems in controlled environments, and they are not very practical for real daily life. People in Sri Lanka change their eating habits a lot because of cultural and religious occasions. This indicates that systems that are based on diet could not work very well. Digital twins and federated learning are two new technologies that look very promising since they help consumers get personalized care while yet protecting their privacy. Most of the studies that are already out there just focus at one part of the problem, like how to keep track of what you eat or how to guess your blood sugar. There aren't any complete systems that combine all of the features into one simple platform.

In the future, Sri Lanka and places like it need to make culturally relevant food identification datasets a top priority. This will immediately improve the accuracy of classifying local foods. The technology used should be easy to access, thus instruments like smart AR glasses that are portable and easy to use should be preferred to help with everyday use. In general, AI models must be designed to protect user privacy, keep them safe, and keep working well in a variety of internet settings. In the end, a design that puts people first and takes into account their literacy levels, cultural views on food, and how well it works with local healthcare systems will determine success.

6 Conclusion

In short, AI gives people with diabetes more interesting ways to help them by giving them individualized food and activity suggestions in a safer and more informative way. Technologies that can tell what food is, guess how much to consume, and keep an eye on your health have come a long way. People may now use them on their smart-watches, smartphones, and smart glasses, which are all common devices. But these tools need to be changed to fit the way people in Sri Lanka eat and live so that they will work there. The next significant step is to stop utilizing a lot of different technologies and start using one simple solution that works for everyone. By designing a tool that is easy to use and culturally smart, we can help people manage their diabetes with more confidence every day.

References

- [1] R. P. Rannan-Eliya *et al.*, "Prevalence of diabetes and pre-diabetes in Sri Lanka: a new global hotspot—estimates from the Sri Lanka Health and Ageing Survey 2018/2019," *BMJ Open Diabetes Res. Care*, vol. 11, no. 1, p. e003160, Feb. 2023, doi: 10.1136/bmjdr-2022-003160.
- [2] P. Katulanda *et al.*, "Prevalence of Diabetes and Pre-Diabetes among Adults in the Western Province of Sri Lanka," *Indian J. Endocrinol. Metab.*, vol. 29, no. 2, pp. 209–216, Mar. 2025, doi: 10.4103/ijem.ijem_383_24.

- [3] M. H. Arambewela *et al.*, “Young-Onset Diabetes in Sri Lanka: Experience From the Developing World,” *J. Diabetes Res.*, vol. 2024, no. 1, p. 7557153, Jan. 2024, doi: 10.1155/jdr/7557153.
- [4] R. Bianco *et al.*, “2D Prediction of the Nutritional Composition of Dishes from Food Images: Deep Learning Algorithm Selection and Data Curation Beyond the Nutrition5k Project,” *Nutrients*, vol. 17, no. 13, p. 2196, Jun. 2025, doi: 10.3390/nu17132196.
- [5] D. Lee and S. Kwon, “Amount Estimation Method for Food Intake Based on Color and Depth Images through Deep Learning,” *Sensors*, vol. 24, no. 7, p. 2044, Mar. 2024, doi: 10.3390/s24072044.
- [6] B. Gonzalez, G. Garcia, S. A. Velastin, H. GholamHosseini, L. Tejeda, and G. Farias, “Automated Food Weight and Content Estimation Using Computer Vision and AI Algorithms,” *Sensors*, vol. 24, no. 23, p. 7660, Nov. 2024, doi: 10.3390/s24237660.
- [7] S.-T. Cheng, Y.-J. Lyu, and C. Teng, “Image-Based Nutritional Advisory System: Employing Multimodal Deep Learning for Food Classification and Nutritional Analysis,” *Appl. Sci.*, vol. 15, no. 9, p. 4911, Apr. 2025, doi: 10.3390/app15094911.
- [8] Z. Revesai and O. P. Kogeda, “Lightweight Interpretable Deep Learning Model for Nutrient Analysis in Mobile Health Applications,” Mar. 14, 2025, *Computer Science and Mathematics*. doi: <https://doi.org/10.3390/digital5020023>.
- [9] I. Papathanail *et al.*, “Evaluation of a Novel Artificial Intelligence System to Monitor and Assess Energy and Macronutrient Intake in Hospitalised Older Patients,” *Nutrients*, vol. 13, no. 12, p. 4539, Dec. 2021, doi: 10.3390/nu13124539.
- [10] A. Mariappan *et al.*, “Personal dietary assessment using mobile devices,” presented at the IS&T/SPIE Electronic Imaging, C. A. Bouman, E. L. Miller, and I. Pollak, Eds., San Jose, CA, Feb. 2009, p. 72460Z. doi: 10.1117/12.813556.
- [11] W. Jia *et al.*, “A Novel Approach to Dining Bowl Reconstruction for Image-Based Food Volume Estimation,” *Sensors*, vol. 22, no. 4, p. 1493, Feb. 2022, doi: 10.3390/s22041493.
- [12] G. Vinod, J. He, Z. Shao, and F. Zhu, “Food Portion Estimation via 3D Object Scaling,” Oct. 10, 2024, *arXiv: arXiv:2404.12257*. doi: 10.48550/arXiv.2404.12257.
- [13] Z. Shao *et al.*, “Towards Learning Food Portion From Monocular Images With Cross-Domain Feature Adaptation,” Mar. 12, 2021, *arXiv: arXiv:2103.07562*. doi: 10.48550/arXiv.2103.07562.
- [14] M. C. Lam, N. A. Suwadi, A. H. Mohd Zainul Arifien, B. K. Poh, N. S. Safii, and J. E. Wong, “An evaluation of a virtual atlas of portion sizes (VAPS) mobile augmented reality for portion size estimation,” *Virtual Real.*, vol. 25, no. 3, pp. 695–707, Sep. 2021, doi: 10.1007/s10055-020-00484-0.
- [15] V. Pini, V. Orso, P. Pluchino, and L. Gamberini, “Augmented grocery shopping: fostering healthier food purchases through AR,” *Virtual Real.*, vol. 27, no. 3, pp. 2117–2128, Sep. 2023, doi: 10.1007/s10055-023-00792-1.
- [16] V. Maksimenko *et al.*, “Do people need guidance to estimate a food portion size? Evidence from an exploratory eye-tracker study,” *Food Qual. Prefer.*, vol. 120, p. 105250, Nov. 2024, doi: 10.1016/j.foodqual.2024.105250.

- [17] I. Mellos and Y. Probst, "Evaluating augmented reality for 'real life' teaching of food portion concepts," *J. Hum. Nutr. Diet.*, vol. 35, no. 6, pp. 1245–1254, Dec. 2022, doi: 10.1111/jhn.13016.
- [18] A. Leone, G. Rescio, G. Diraco, A. Manni, P. Siciliano, and A. Caroppo, "Ambient and Wearable Sensor Technologies for Energy Expenditure Quantification of Ageing Adults," *Sensors*, vol. 22, no. 13, p. 4893, Jun. 2022, doi: 10.3390/s22134893.
- [19] A. D. Zahedani *et al.*, "Digital health application integrating wearable data and behavioral patterns improves metabolic health," *Npj Digit. Med.*, vol. 6, no. 1, p. 216, Nov. 2023, doi: 10.1038/s41746-023-00956-y.
- [20] D. Adams and E. Nsugbe, "Predictive Glucose Monitoring for People with Diabetes Using Wearable Sensors," in *The 8th International Electronic Conference on Sensors and Applications*, MDPI, Nov. 2021, p. 20. doi: 10.3390/ecsa-8-11317.
- [21] M. Shi, J. Zhou, and M. Cai, "Multiple physiological and behavioural parameters identification for dietary monitoring using wearable sensors: a study protocol," *BMC Nutr.*, vol. 11, no. 1, p. 183, Oct. 2025, doi: 10.1186/s40795-025-01168-1.
- [22] S. M. Dimitratos, J. B. German, and S. E. Schaefer, "Wearable Technology to Quantify the Nutritional Intake of Adults: Validation Study," *JMIR MHealth UHealth*, vol. 8, no. 7, p. e16405, Jul. 2020, doi: 10.2196/16405.
- [23] F. Prendin, S. Del Favero, M. Vettoretti, G. Sparacino, and A. Facchinetti, "Forecasting of Glucose Levels and Hypoglycemic Events: Head-to-Head Comparison of Linear and Nonlinear Data-Driven Algorithms Based on Continuous Glucose Monitoring Data Only," *Sensors*, vol. 21, no. 5, p. 1647, Feb. 2021, doi: 10.3390/s21051647.
- [24] S. Langerica *et al.*, "Deep Learning-Based Glucose Prediction Models: A Guide for Practitioners and a Curated Dataset for Improved Diabetes Management," *IEEE Open J. Eng. Med. Biol.*, vol. 5, pp. 467–475, 2024, doi: 10.1109/OJEMB.2024.3365290.
- [25] T. S. Karunaratna and Z. Liang, "Development of Non-Invasive Continuous Glucose Prediction Models Using Multi-Modal Wearable Sensors in Free-Living Conditions," *Sensors*, vol. 25, no. 10, p. 3207, May 2025, doi: 10.3390/s25103207.
- [26] S. Rancati *et al.*, "Exploration of Foundational Models for Blood Glucose Forecasting in Type-1 Diabetes Pediatric Patients," *Diabetology*, vol. 5, no. 6, pp. 584–599, Nov. 2024, doi: 10.3390/diabetology5060042.
- [27] I. Rodríguez-Rodríguez, I. Chatzigiannakis, J.-V. Rodríguez, M. Maranghi, M. Gentili, and M.-Á. Zamora-Izquierdo, "Utility of Big Data in Predicting Short-Term Blood Glucose Levels in Type 1 Diabetes Mellitus Through Machine Learning Techniques," *Sensors*, vol. 19, no. 20, p. 4482, Oct. 2019, doi: 10.3390/s19204482.
- [28] Z. Johnson and M. J. Saikia, "Digital Twins for Healthcare Using Wearables," *Bioengineering*, vol. 11, no. 6, p. 606, Jun. 2024, doi: 10.3390/bioengineering11060606.
- [29] M. N. Kamel Boulos and P. Zhang, "Digital Twins: From Personalised Medicine to Precision Public Health," *J. Pers. Med.*, vol. 11, no. 8, p. 745, Jul. 2021, doi: 10.3390/jpm11080745.
- [30] P. Shamanna *et al.*, "Personalized nutrition in type 2 diabetes remission: application of digital twin technology for predictive glycemic control," *Front. Endocrinol.*, vol. 15, p. 1485464, Nov. 2024, doi: 10.3389/fendo.2024.1485464.

- [31] K. Zhang *et al.*, “Concepts and applications of digital twins in healthcare and medicine,” *Patterns*, vol. 5, no. 8, p. 101028, Aug. 2024, doi: 10.1016/j.patter.2024.101028.
- [32] H. Zhao, D. Sui, Y. Wang, L. Ma, and L. Wang, “Privacy-Preserving Federated Learning Framework for Multi-Source Electronic Health Records Prognosis Prediction,” *Sensors*, vol. 25, no. 8, p. 2374, Apr. 2025, doi: 10.3390/s25082374.
- [33] R. Haripriya, N. Khare, and M. Pandey, “Privacy-preserving federated learning for collaborative medical data mining in multi-institutional settings,” *Sci. Rep.*, vol. 15, no. 1, p. 12482, Apr. 2025, doi: 10.1038/s41598-025-97565-4.
- [34] K. A. Nfor, T. P. Theodore Armand, K. P. Ismaylova, M.-I. Joo, and H.-C. Kim, “An Explainable CNN and Vision Transformer-Based Approach for Real-Time Food Recognition,” *Nutrients*, vol. 17, no. 2, p. 362, Jan. 2025, doi: 10.3390/nu17020362.
- [35] Y. Han, Q. Cheng, W. Wu, and Z. Huang, “DPF-Nutrition: Food Nutrition Estimation via Depth Prediction and Fusion,” *Foods*, vol. 12, no. 23, p. 4293, Nov. 2023, doi: 10.3390/foods12234293.
- [36] L. Bossard, M. Guillaumin, and L. Van Gool, “Food-101 – Mining Discriminative Components with Random Forests,” in *Computer Vision – ECCV 2014*, vol. 8694, D. Fleet, T. Pajdla, B. Schiele, and T. Tuytelaars, Eds., in Lecture Notes in Computer Science, vol. 8694, Cham: Springer International Publishing, 2014, pp. 446–461. doi: 10.1007/978-3-319-10599-4_29.
- [37] Z. Chen, J. Wang, and Y. Wang, “Enhancing Food Image Recognition by Multi-Level Fusion and the Attention Mechanism,” *Foods*, vol. 14, no. 3, p. 461, Jan. 2025, doi: 10.3390/foods14030461.
- [38] L. Boyd, N. Nnamoko, and R. Lopes, “Fine-Grained Food Image Recognition: A Study on Optimising Convolutional Neural Networks for Improved Performance,” *J. Imaging*, vol. 10, no. 6, p. 126, May 2024, doi: 10.3390/jimaging10060126.